

Augmented stability conditions

Antonios-Alexandros Robotis

Based on joint works with Daniel Halpern-Leistner and Jeffrey Jiang

Derived categories

Throughout: X is a complex projective manifold.

- $\mathbf{D}^b(X)$ = bounded derived category of coherent sheaves of X
- $\mathbf{D}^b(X)$ is an interesting and subtle invariant of X , from which some classical invariants of X can be extracted:
 - 1 $K_0(X)$ = Grothendieck group of alg. vector bundles on X
 - 2 $H^{\text{even}}(X; \mathbf{Q})$ and $H^{\text{odd}}(X; \mathbf{Q})$
 - 3 Certain sums of Hodge numbers of X
- Derived equivalence vs birational equivalence?
 - 1 for X with ω_X (anti)ample, $X \cong Y \iff \mathbf{D}^b(X) \simeq \mathbf{D}^b(Y)$
 - 2 $\mathbf{P}^1 \times \mathbf{P}^1 \simeq^{\text{bir}} \mathbf{P}^2$ but $\mathbf{D}^b(\mathbf{P}^1 \times \mathbf{P}^1) \not\simeq \mathbf{D}^b(\mathbf{P}^2)$
 - 3 A and $A^\vee$ not birational in general, but $\mathbf{D}^b(A) \simeq \mathbf{D}^b(A^\vee)$
 - 4 Conjecture: (D-equivalence) birational CY varieties have equivalent derived categories

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Semiorthogonal decompositions

Definition

A *semiorthogonal decomposition* of a Δ' d category $\mathcal{D}$ is a collection of full additive subcategories $\mathcal{D}_1, \dots, \mathcal{D}_k$ such that

- 1 $\mathcal{D}_1, \dots, \mathcal{D}_k$ generate $\mathcal{D}$
- 2 $i < j$ implies $\mathrm{Hom}_{\mathcal{D}}(\mathcal{D}_j, \mathcal{D}_i) = 0$.

Written $\mathcal{D} = \langle \mathcal{D}_1, \dots, \mathcal{D}_k \rangle$

$\approx$ analogue of split filtration or direct sum decomp.

Example (Beilinson)

Let $\mathcal{D} = \mathbf{D}^b(\mathbf{P}^n)$ and let $\mathcal{D}_i = \langle \mathcal{O}(i) \rangle$ for $i = 0, \dots, n$. The set $\{\mathcal{O}, \dots, \mathcal{O}(n)\}$ is called a *full exceptional collection*. Gives

$$K_0(\mathbf{P}^n) = \bigoplus_{i=0}^n \mathbf{Z} \cdot [\mathcal{O}(i)].$$

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Bridgeland stability

$\text{Stab}(\mathcal{D}) = \text{Stab}(X)$ – space of *stability conditions* $(Z, \mathcal{P})$ on $\mathbf{D}^b(X)$

- *central charge*: $Z \in \text{Hom}(K_0(X), \mathbf{C})$, which factors through $\text{ch} : K_0(X) \rightarrow H_{\text{alg}}^*(X; \mathbf{C})$
- $\mathcal{P}$ is a *slicing*, i.e. a collection of full additive subcategories $\{\mathcal{P}(\phi)\}_{\phi \in \mathbf{R}}$ + axioms, e.g. $\psi > \phi \Rightarrow \text{Hom}_{\mathcal{D}}(\mathcal{P}(\psi), \mathcal{P}(\phi)) = 0$
- $\mathcal{P}(\phi) :=$ semistable objects of phase ϕ ,

$$\forall E \in \mathcal{P}(\phi), \quad Z(E) \in \mathbf{R}_{>0} \cdot \exp(i\pi\phi).$$

- (*Bridgeland*) $\text{Stab}(X) \rightarrow \text{Hom}(H_{\text{alg}}^*(X), \mathbf{C})$ given by $(Z, \mathcal{P}) \mapsto Z$ is a local homeo. So, $\text{Stab}(X)$ is a $\mathbf{C}$ -manifold modeled on $H_{\text{alg}}^*(X; \mathbf{C})^\vee$.
- Natural $\mathbf{C}$ -action on $\text{Stab}(X)$: $w \cdot (Z, \mathcal{P}) = (e^w \cdot Z, \mathcal{P}^w)$ where $\mathcal{P}^w(\phi) = \mathcal{P}(\phi - \Im(w)/\pi)$.

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$\text{Stab}(\mathcal{D}) = \text{Stab}(X)$ – space of *stability conditions* $(Z, \mathcal{P})$ on $\mathbf{D}^b(X)$

- *central charge*: $Z \in \text{Hom}(K_0(X), \mathbf{C})$, which factors through $\text{ch} : K_0(X) \rightarrow H_{\text{alg}}^*(X; \mathbf{C})$
- $\mathcal{P}$ is a *slicing*, i.e. a collection of full additive subcategories $\{\mathcal{P}(\phi)\}_{\phi \in \mathbf{R}}$ + axioms, e.g. $\psi > \phi \Rightarrow \text{Hom}_{\mathcal{D}}(\mathcal{P}(\psi), \mathcal{P}(\phi)) = 0$
- $\mathcal{P}(\phi) :=$ semistable objects of phase ϕ ,

$$\forall E \in \mathcal{P}(\phi), \quad Z(E) \in \mathbf{R}_{>0} \cdot \exp(i\pi\phi).$$

- (*Bridgeland*) $\text{Stab}(X) \rightarrow \text{Hom}(H_{\text{alg}}^*(X), \mathbf{C})$ given by $(Z, \mathcal{P}) \mapsto Z$ is a local homeo. So, $\text{Stab}(X)$ is a $\mathbf{C}$ -manifold modeled on $H_{\text{alg}}^*(X; \mathbf{C})^\vee$.
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Stability and SODs interact

Definition

A *polarized* SOD of $\mathcal{D}$ is $\mathcal{D} = \langle \mathcal{D}_1, \dots, \mathcal{D}_k \rangle$ equipped with a choice of $\sigma_i \in \text{Stab}(\mathcal{D}_i)$ for $i = 1, \dots, k$.

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Proposition (CP '2010, HLJR '2023)

Given a polarized SOD $\mathbf{D}^b(X) = \langle \mathcal{D}_1, \dots, \mathcal{D}_k | \sigma_1, \dots, \sigma_k \rangle$, $\exists$ $\langle \mathcal{D}_1, \dots, \mathcal{D}_k | \sigma'_1, \dots, \sigma'_k \rangle$ and $\sigma = (Z, \mathcal{P}) \in \text{Stab}(\mathcal{D})$ glued from $(\sigma'_1, \dots, \sigma'_k)$: that is,

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The motivating problem

In arXiv:2301.13168, Halpern-Leistner proposes noncommutative minimal model program (NMMP)

Heuristic

Given $\sigma_0 = (Z_0, \mathcal{P}_0) \in \text{Stab}(X)$, solving “canonical ODEs” in $H_{\text{alg}}^*(X; \mathbf{C})^\vee$ with starting point Z_0 gives paths $Z_t : [0, \infty) \rightarrow H_{\text{alg}}^*(X; \mathbf{C})^\vee$ which lift to $\sigma_t : [0, \infty) \rightarrow \text{Stab}(X)$.

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A generic quasi-convergent path σ_t gives a semiorthogonal decomposition $\mathcal{D} = \langle \mathcal{D}_1, \dots, \mathcal{D}_n \rangle$ plus $\sigma_i \in \text{Stab}(\mathcal{D}_i) / \mathbb{C}$ for $i = 1, \dots, n$.

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Quasi-convergent paths, cont.

There are enough quasi-convergent paths!

Theorem (HLJR '23)

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The proof uses the gluing construction of Collins - Polishchuk

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Qc. paths should converge in a (partial) compactification of $\text{Stab}(\mathcal{D}) / \mathbb{C}$ to boundary points which correspond to polarised SODs (+ some additional data!)

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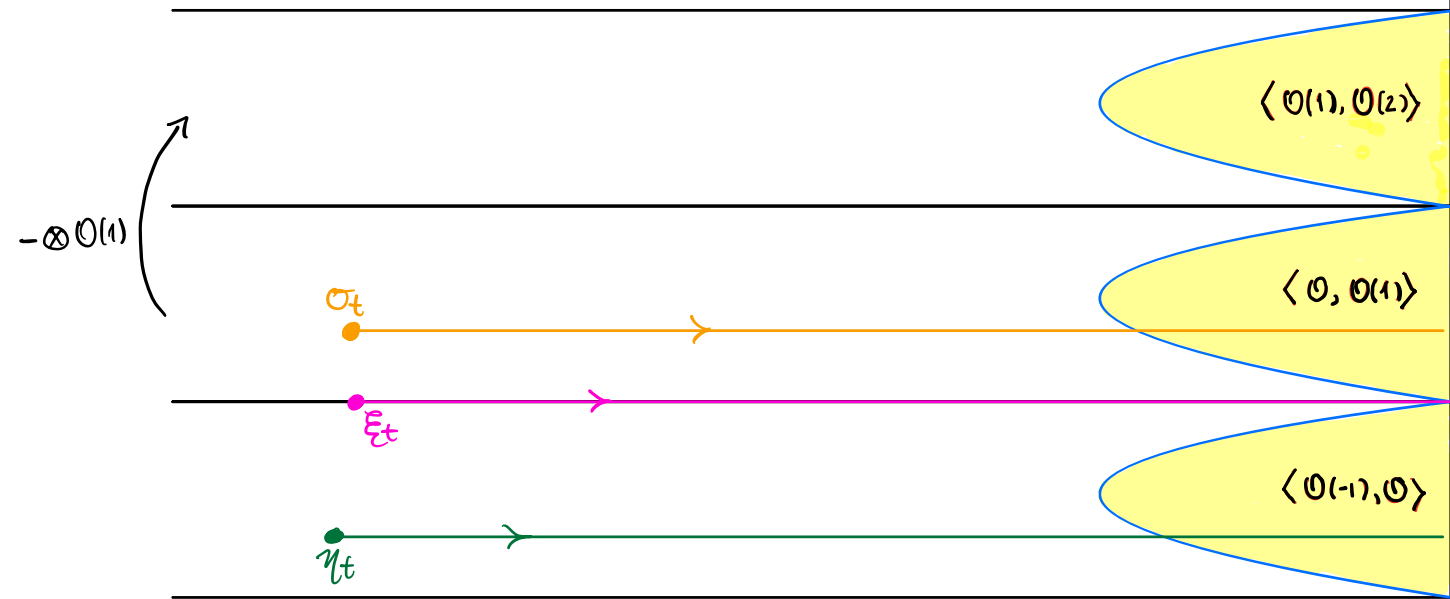
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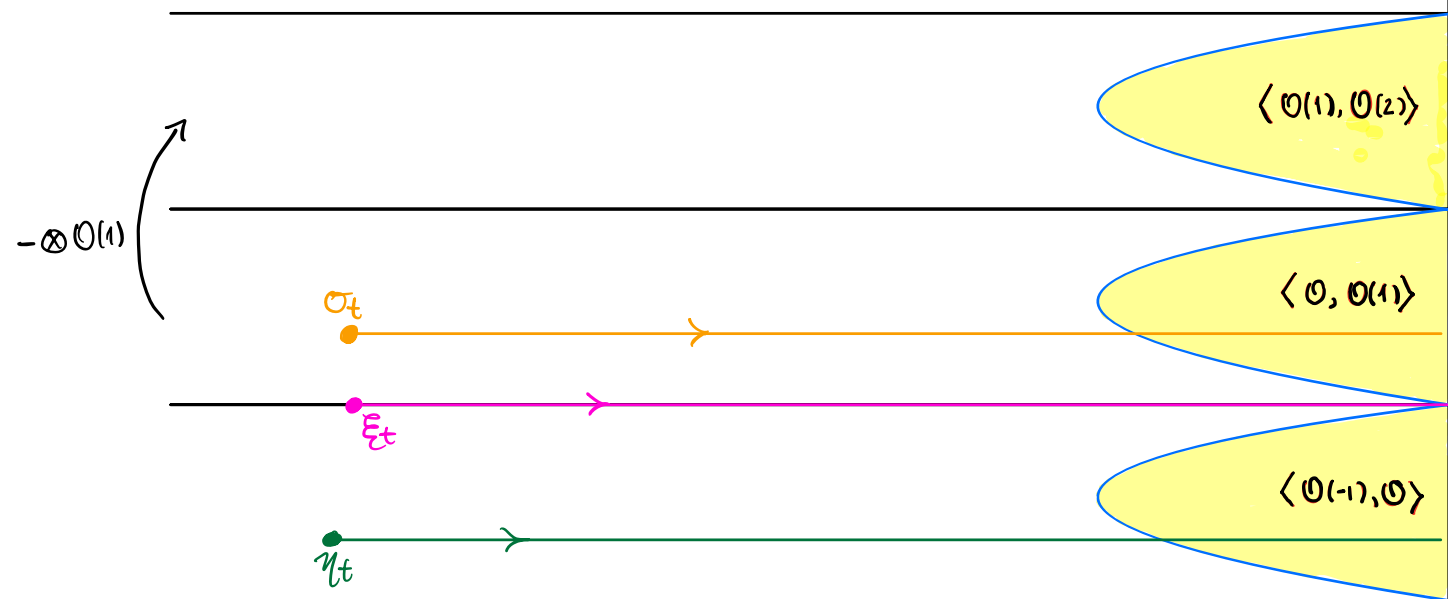
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$$\sigma_t \xrightarrow{\text{Thm.}} \langle \mathcal{O}, \mathcal{O}(1) \rangle$$

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$$\xi_t \xrightarrow{\quad} ???$$

Coordinates on the stability manifold

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Proof: $\mu \in \text{Aut}(\mathbf{P}^1)$: $\mu(\infty) = \infty \Rightarrow \mu(z) = az + b$ and $\mu^*(dz) = dz$ implies $a = 1$.

We compactify $\mathbf{C}^n / \mathbf{C}$ by introducing a “new” moduli space of marked genus 0 curves, called multiscale lines – inspired by multiscale differentials of Bainbridge - Chen - Gendron - Grushevsky - Möller '19.

Note: dz on $\mathbf{P}^1$ is characterised up to a scalar as a meromorphic differential with an order 2 pole at ∞ ; i.e, $dz \in \Gamma(\Omega_{\mathbf{P}^1}(2\infty)) = \mathbf{C}$.

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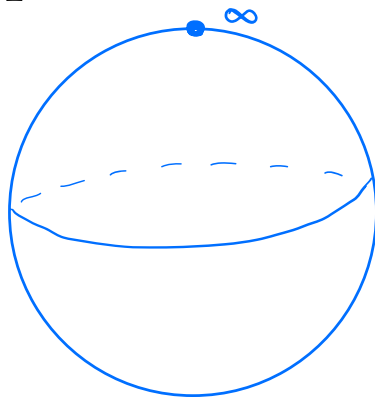
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Proof: $\mu \in \text{Aut}(\mathbf{P}^1)$: $\mu(\infty) = \infty \Rightarrow \mu(z) = az + b$ and $\mu^*(dz) = dz$ implies $a = 1$.

We compactify $\mathbf{C}^n / \mathbf{C}$ by introducing a “new” moduli space of marked genus 0 curves, called multiscale lines – inspired by multiscale differentials of Bainbridge - Chen - Gendron - Grushevsky - Möller '19.

Note: dz on $\mathbf{P}^1$ is characterised up to a scalar as a meromorphic differential with an order 2 pole at ∞ ; i.e, $dz \in \Gamma(\Omega_{\mathbf{P}^1}(2\infty)) = \mathbf{C}$.



Multiscale lines

Definition (Part I)

An n -marked *multiscale line* is $(\Sigma, p_\infty, \preceq, \omega_\bullet, p_1, \dots, p_n)$ where

- ① Σ is a nodal genus 0 curve over $\mathbb{C}$
- ② p_∞ is a special “top” point on the curve
- ③ $\preceq$ is a level structure
- ④ $p_1, \dots, p_n$ are marked points which may collide with each other but not with p_∞

+ conditions: e.g. $p_1, \dots, p_n$ are all on components furthest from p_∞ and all such components are at the same “level” ...

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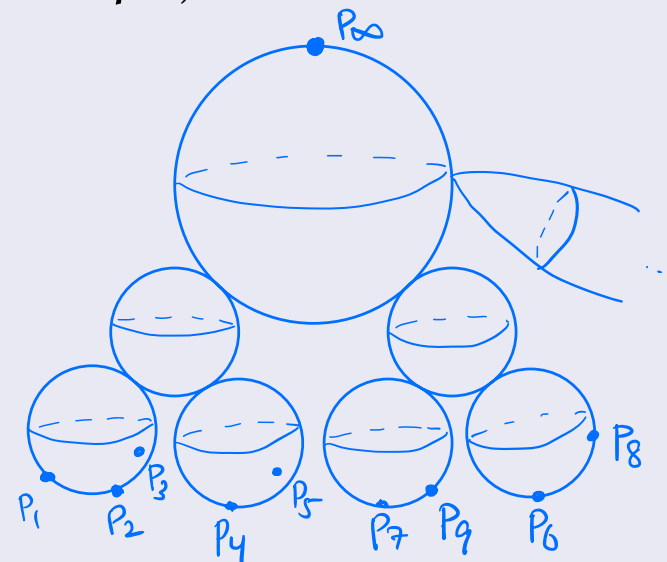
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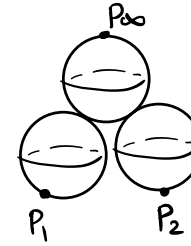
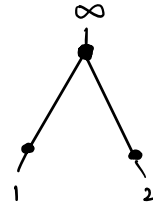
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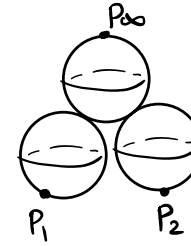
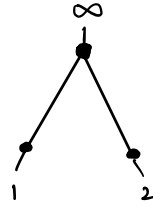
Some combinatorial types:

$n = 2$.

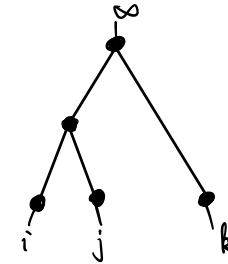
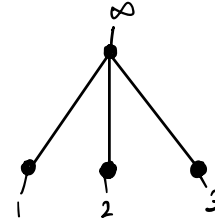
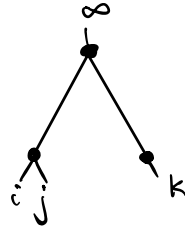


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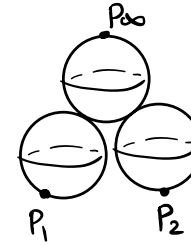
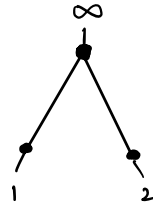


$n = 3$.

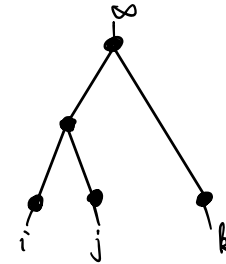
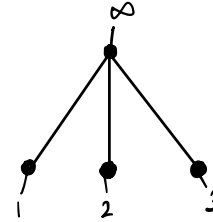
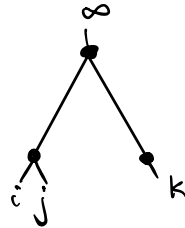


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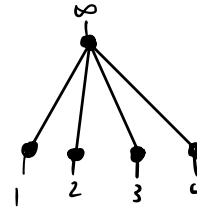
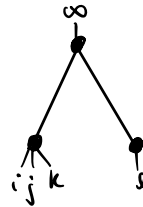
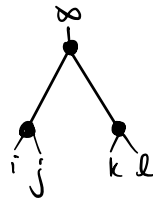
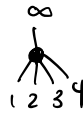
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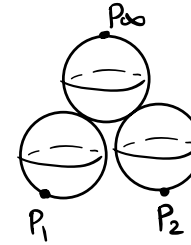
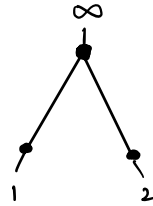


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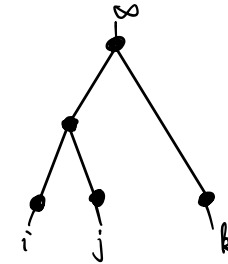
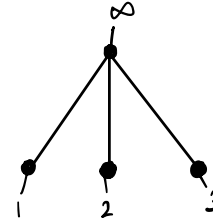
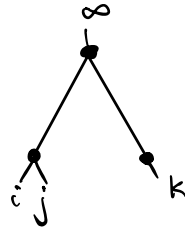


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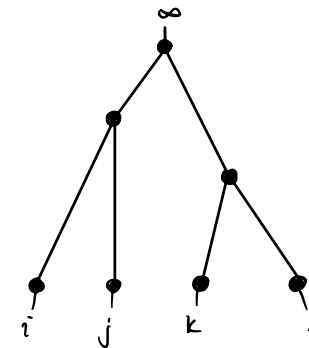
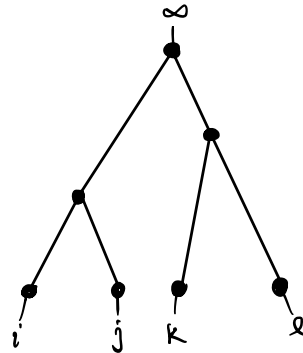
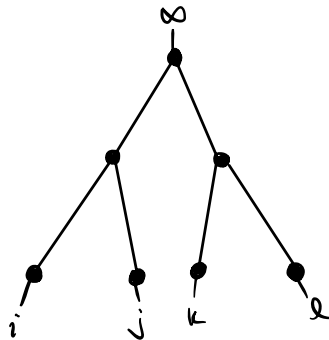
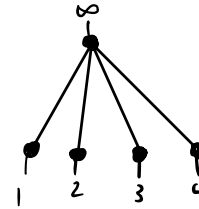
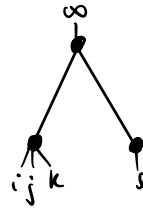
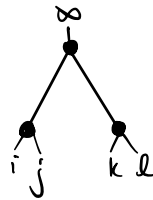
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Multiscale lines

Definition (Part II)

... and $\omega_{\bullet} = \{\omega_v\}_{v \in V(\Sigma)} \rightsquigarrow$ meromorphic 1-forms, one per component Σ_v , with unique order 2 pole at the node connecting Σ_v to the root (resp. at p_{∞} if $v = \text{root}$)

When $\Sigma = \mathbf{P}^1$, a multiscale line is just $(\mathbf{P}^1, \infty, \omega = \lambda dz, p_1, \dots, p_n)$ for $\lambda \in \mathbf{C}^* \rightsquigarrow \{\text{irred. } n\text{-marked multiscale lines}\} / \sim = \mathbf{C}^n / \mathbf{C}$.

For each $v \in V(\Sigma)$, ω_v on $\Sigma_v \longleftrightarrow \lambda_v \in \mathbf{C}^*$.

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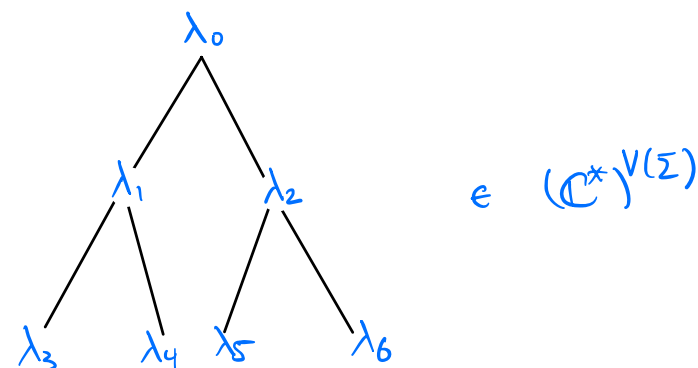
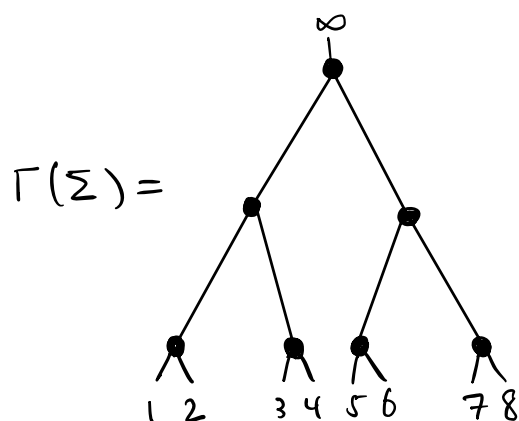
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Definition

A $\mathbf{C}$ -projective (resp. $\mathbf{R}$ -oriented) iso. of multiscale lines $f : \Sigma \rightarrow \Sigma'$ is an iso. of curves that preserves level structures and marked points s.t.

$$f^*(\omega'_v) = c_v \omega_v \quad \text{for } c_v \in \mathbf{C}^* \text{ (resp. } \mathbf{R}_{>0})$$

and $c_v = c_w$ if $v \sim w$ and $c_v = 1$ for v a bottom comp. of Σ .

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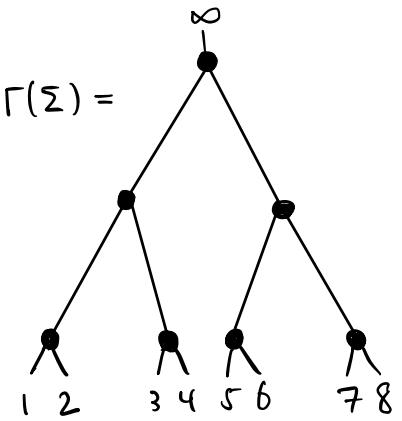
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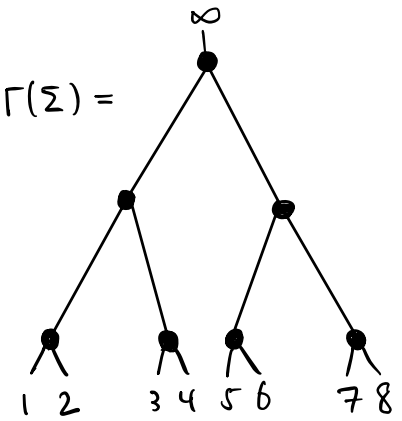
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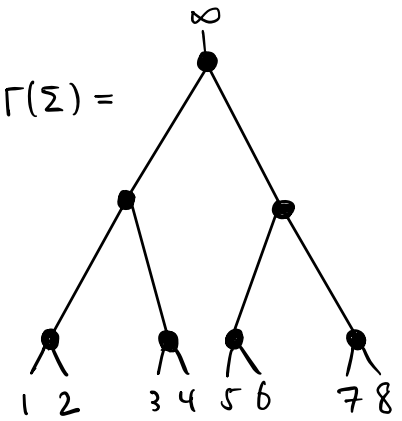
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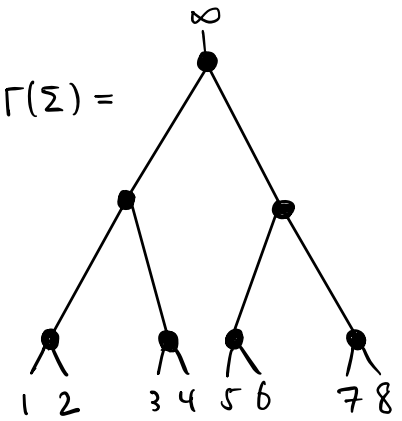
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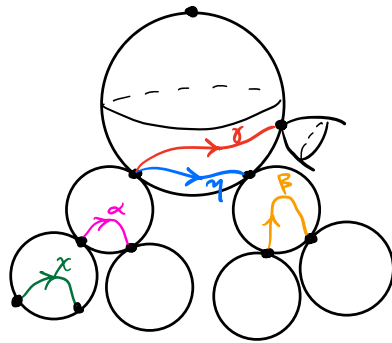
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In the irred.
case, all three
of these notions
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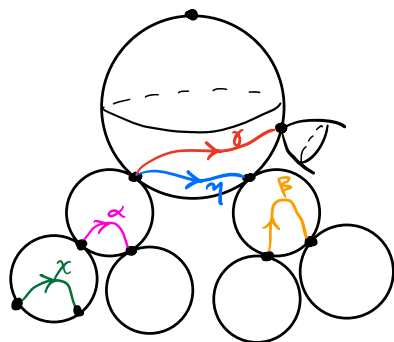
Examples



Up to $\mathbb{C}$ -projective iso.

$$\int_{\tilde{x}} \omega_u, \quad \frac{\int_{\tilde{\gamma}} \omega_{\text{root}}}{\int_{\tilde{\gamma}} \omega_{\text{root}}}, \quad \frac{\int_{\tilde{\beta}} \omega_w}{\int_{\tilde{\alpha}} \omega_v} \quad \text{are defined.}$$

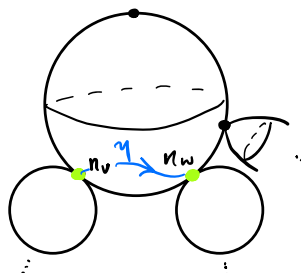
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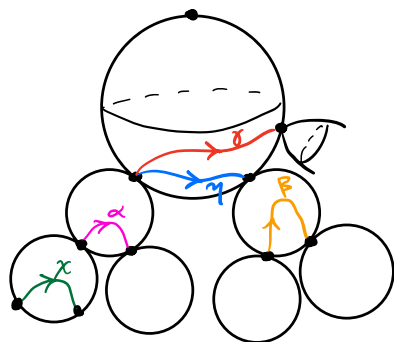
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Up to $\mathbb{R}$ -oriented iso,
more functions are defined:



$$p(v, w) := \frac{\int_{\gamma} \omega_{\text{root}}}{|\int_{\gamma} \omega_{\text{root}}|} \in S^1$$

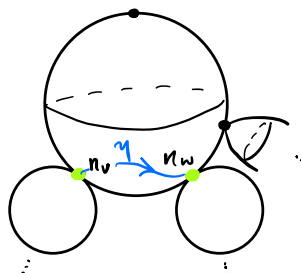
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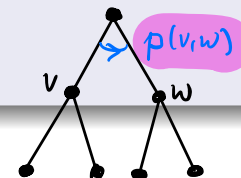
Up to $\mathbb{R}$ -oriented iso.
more functions are defined:



$$p(v, w) := \frac{\int_{\gamma} \omega_{\text{root}}}{|\int_{\gamma} \omega_{\text{root}}|} \in S^1$$

Heuristic

A multiscale line Σ up to real-oriented isomorphism gives its dual tree the structure of a level graph with *angles* between edges



Moduli spaces

- 1 $\mathcal{A}_n := \{\mathbf{C} - \text{proj. iso. classes of } n\text{-marked multiscale lines}\}$
- 2 $\mathbf{C}^n / \mathbf{C} = \mathcal{A}_n^\circ \subset \mathcal{A}_n$ is the set of irreducible multiscale lines
- 3 Coordinates on $\mathcal{A}_n$ are constructed using the integral functions from the last slide.

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$\mathcal{A}_n$ is a compact complex algebraic manifold containing $\mathbf{C}^n / \mathbf{C}$ as an open dense subset. The boundary $D := \mathcal{A}_n \setminus \mathcal{A}_n^\circ$ is snc.

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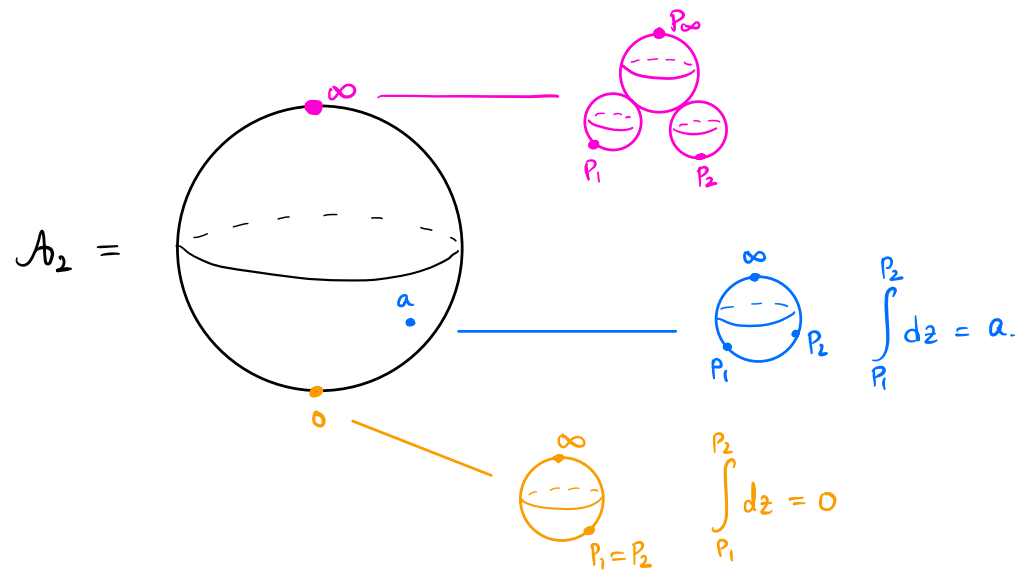
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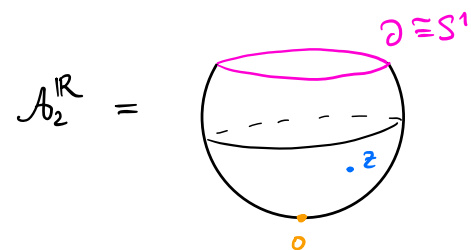
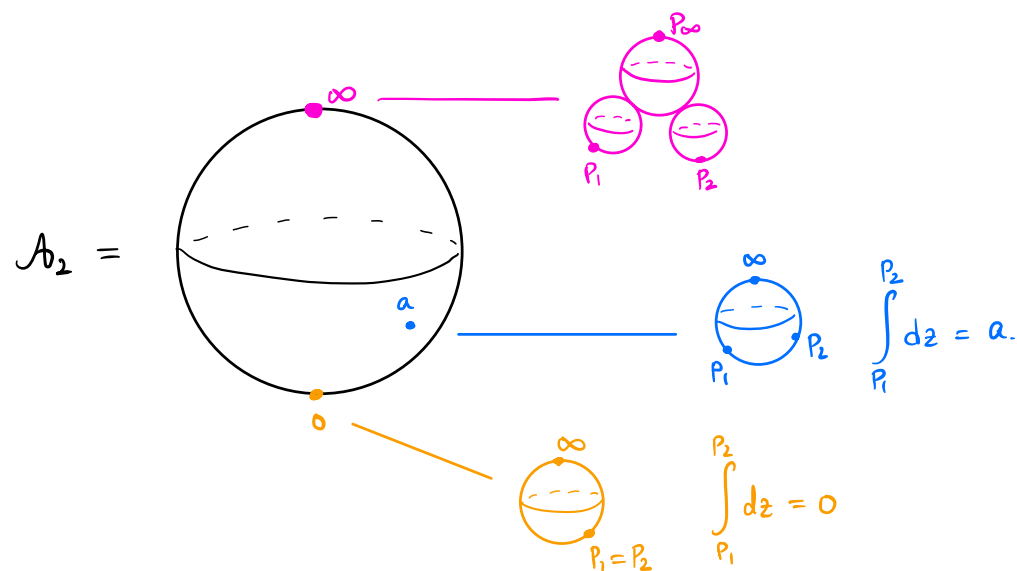
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$\mathcal{A}_2$ and $\mathcal{A}_2^R$



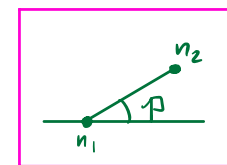
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coordinate on

$$\varphi(p_1, p_2) =$$

$$\frac{\int_{n_1}^{n_2} \omega_{\text{root}}}{\int_{n_1}^{n_2} \omega_{\text{root}}} \in S^1$$



$$\mathbb{C} = \mathbb{P}'_{\text{root}} \setminus \{p_\infty\}$$

Multiscale decompositions

Definition (Multiscale decomposition)

A *multiscale decomposition* $\mathcal{D} = \langle \mathcal{D}_\bullet \rangle_\Sigma$ is:

- ① an un-marked multiscale line $(\Sigma, p_\infty, \preceq, \omega_\bullet)$ and
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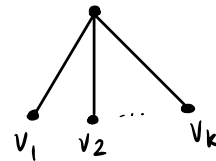
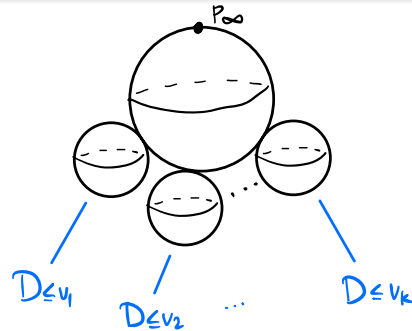
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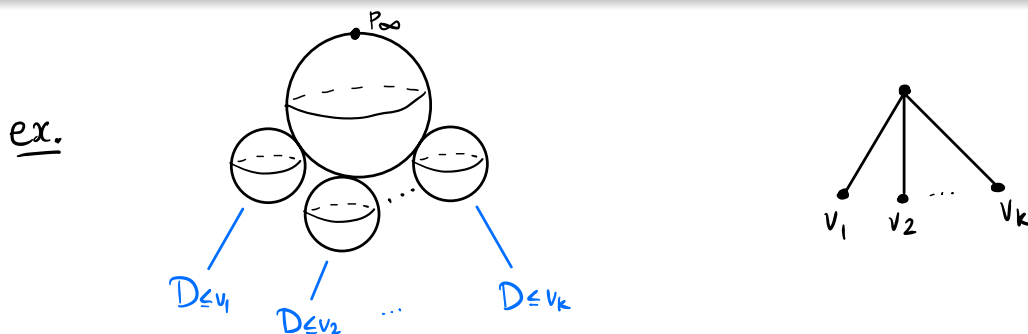


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Heuristic

A multiscale decomposition is a categorical structure interpolating between a *filtration* and a *semiorthogonal decomposition* of $\mathcal{D}$.

When $\Sigma \cong \mathbf{P}^1$, a multiscale decomp. is trivial, just $\mathcal{D} = \langle \mathcal{D} \rangle_{\mathbf{P}^1}$. Next, let

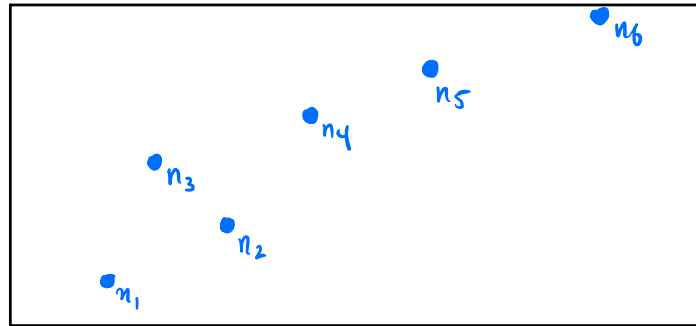
$$\Gamma(\Sigma) = \begin{array}{c} \bullet \\ \swarrow \quad \downarrow \quad \searrow \\ \bullet \quad \bullet \quad \bullet \\ v_1 \quad v_2 \quad v_b \end{array}$$

Consider the following configurations of nodes on $\Sigma_{\text{root}} \setminus p_\infty = \mathbf{C}$:

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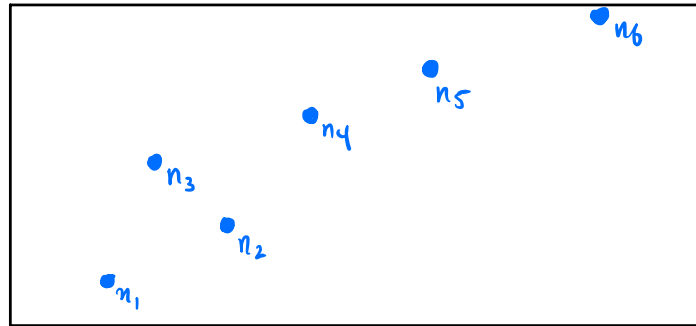


$\Im p(v_i, v_j) > 0 \Rightarrow \text{Hom}(\mathcal{D}_{\leq v_j}, \mathcal{D}_{\leq v_i}) = 0$; get so d $\mathcal{D} = \langle \mathcal{D}_{\leq v_1}, \dots, \mathcal{D}_{\leq v_6} \rangle$.

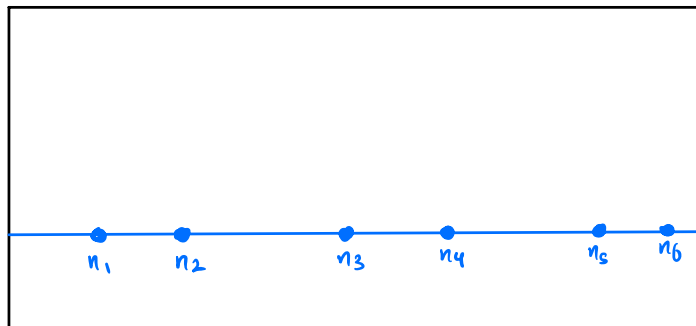
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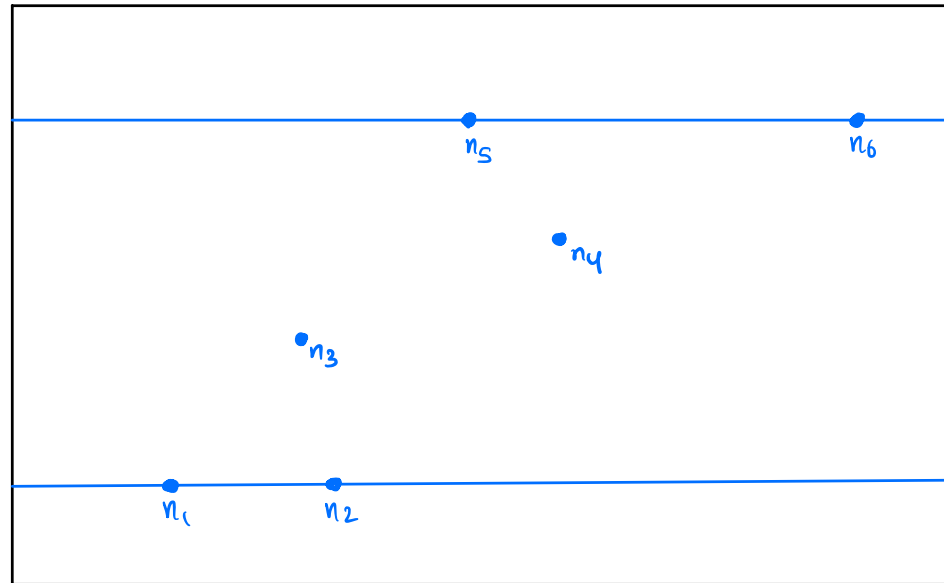


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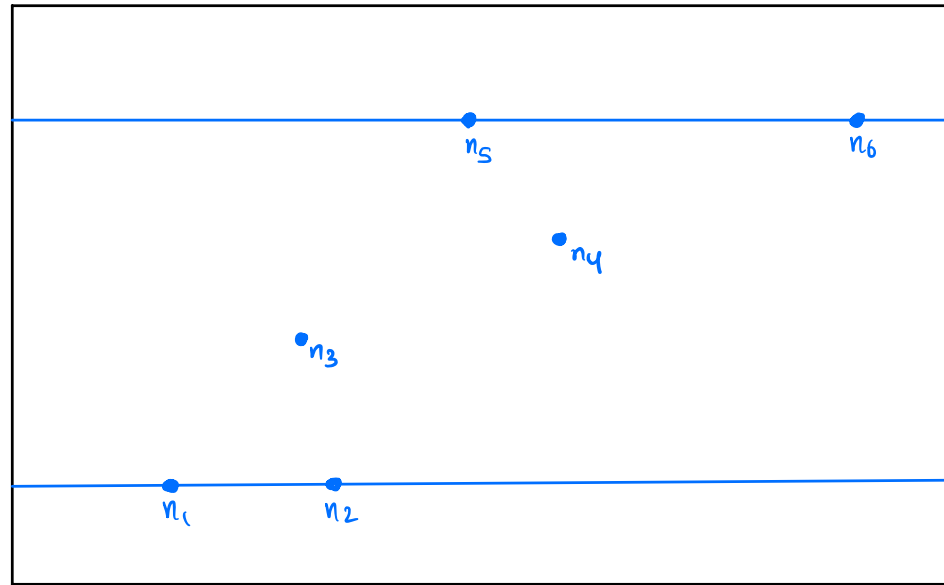


$\mathfrak{p}(v_i, v_j) = 1 \Rightarrow \mathcal{D}_{\leq v_i} \subsetneq \mathcal{D}_{\leq v_j}$; get filt. $0 \subsetneq \mathcal{D}_{\leq v_1} \subsetneq \dots \subsetneq \mathcal{D}_{\leq v_6} = \mathcal{D}$.

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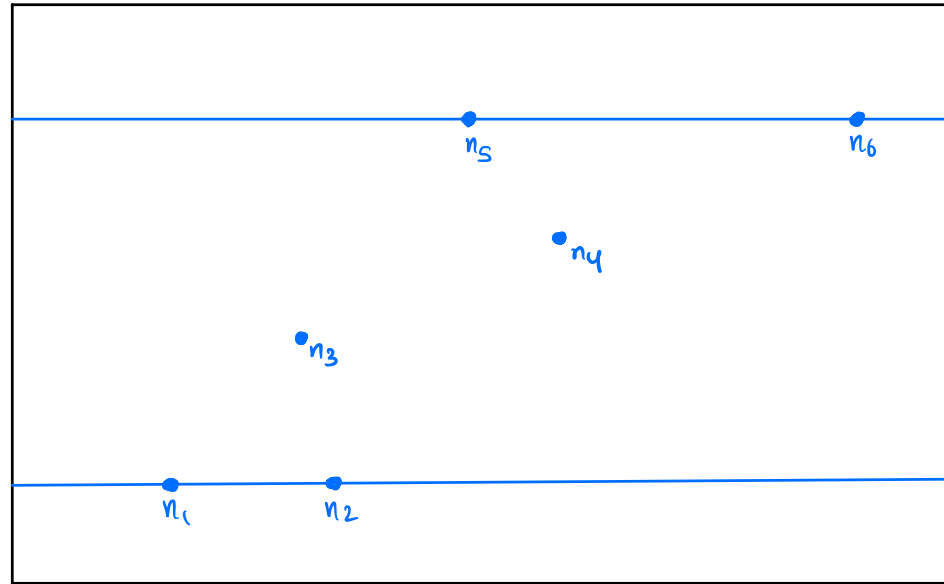
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We get a filtered SOD,

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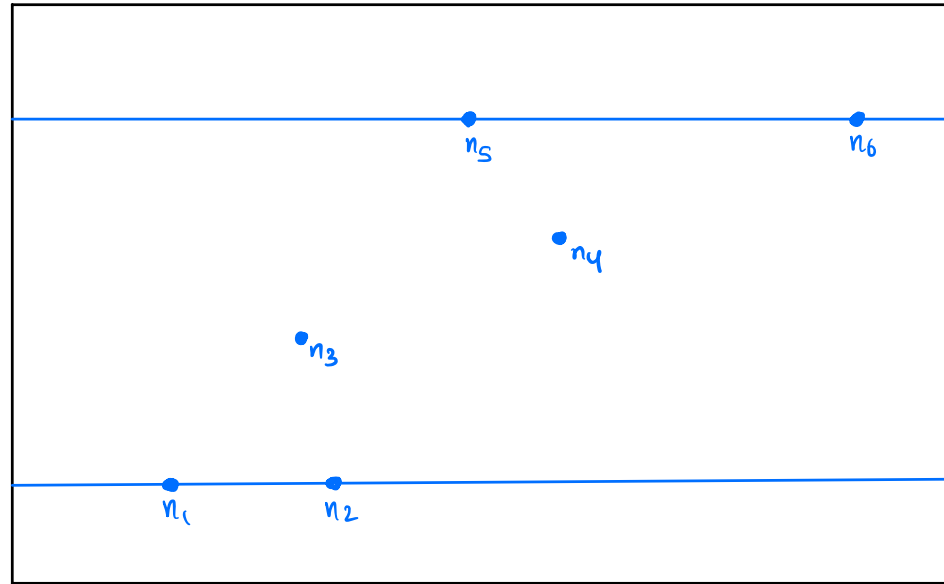


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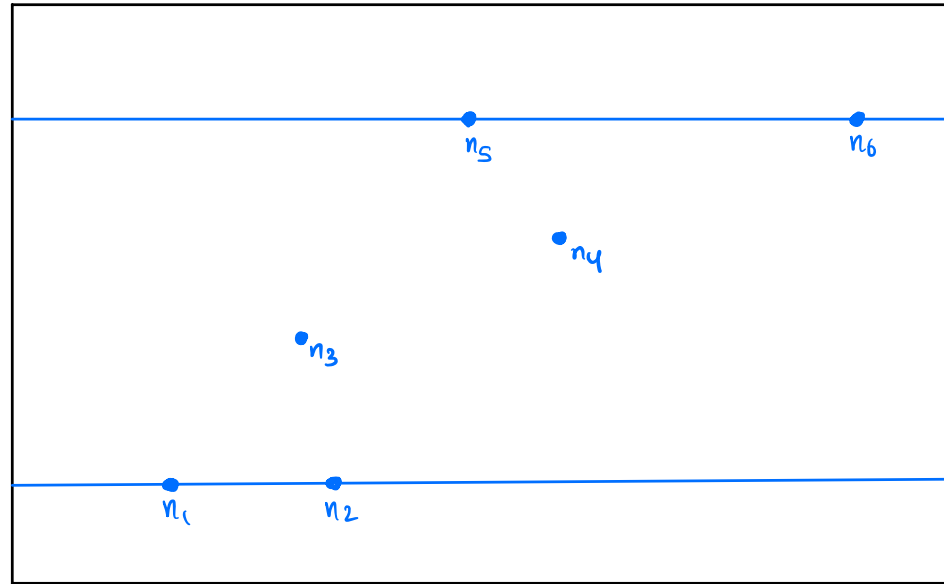
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Augmented stability conditions

Definition (Augmented stability condition)

An *augmented stability condition* $\sigma = \langle \mathcal{D}_\bullet | \sigma_\bullet \rangle_\Sigma$ is a multiscale decomposition $\mathcal{D} = \langle \mathcal{D}_\bullet \rangle_\Sigma$ such that $\text{gr}_v(\mathcal{D}_\bullet)$ is equipped with $\sigma_v \in \text{Stab}(\text{gr}_v(\mathcal{D}_\bullet)) / \mathbf{C}$ for each $v \in V(\Sigma)_{\text{bot}}$.

- ① $\mathcal{A}\text{Stab}(\mathcal{D}) =$ space of augmented stability conditions
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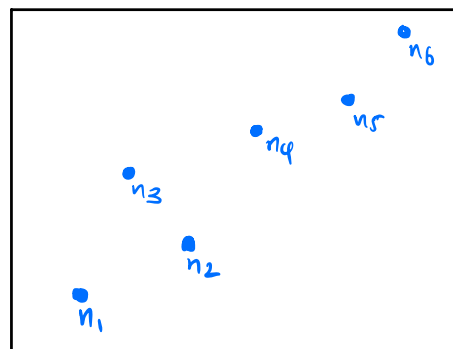
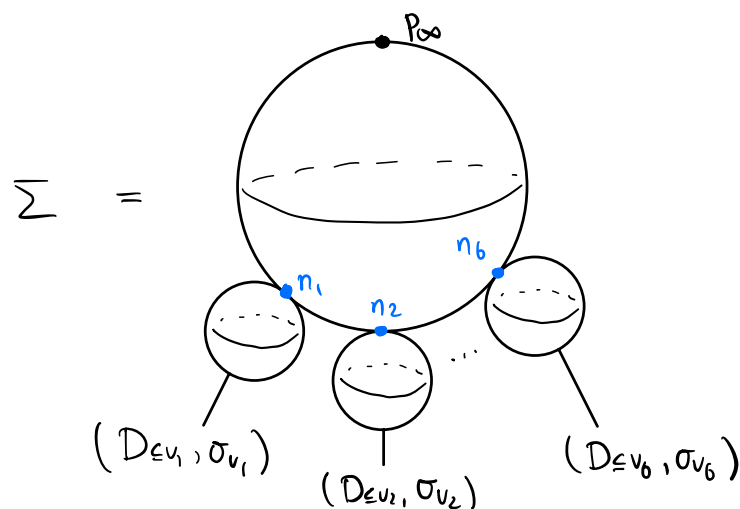
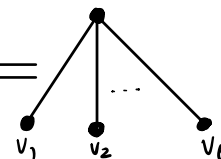
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 (polarized sOD + Σ)

Main Theorem

$\mathcal{A} \text{Stab}(\mathcal{D}) :=$ set of augmented stability conditions.

Theorem (Halpern-Leistner, R.)

There is a Hausdorff topology on $\mathcal{A} \text{Stab}(\mathcal{D})$ such that

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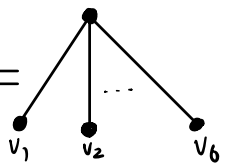
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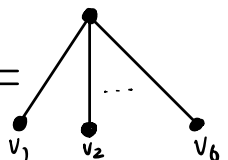
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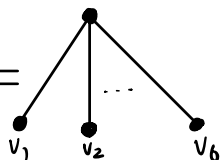
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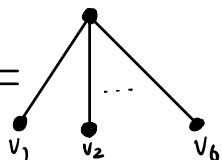
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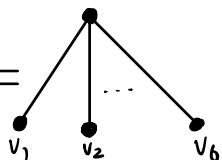
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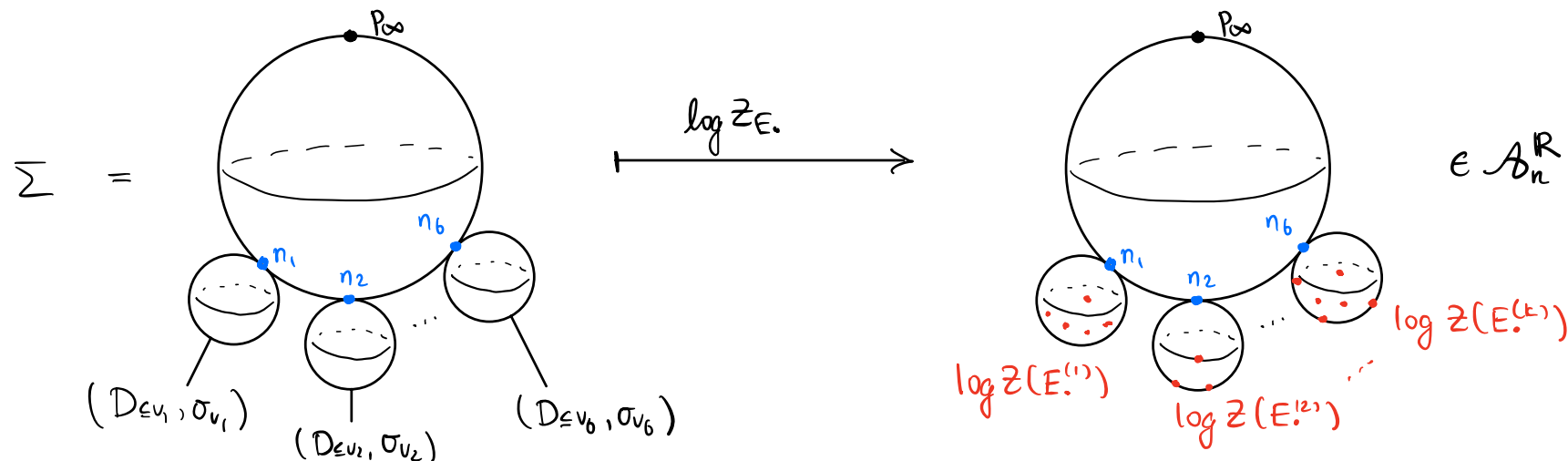
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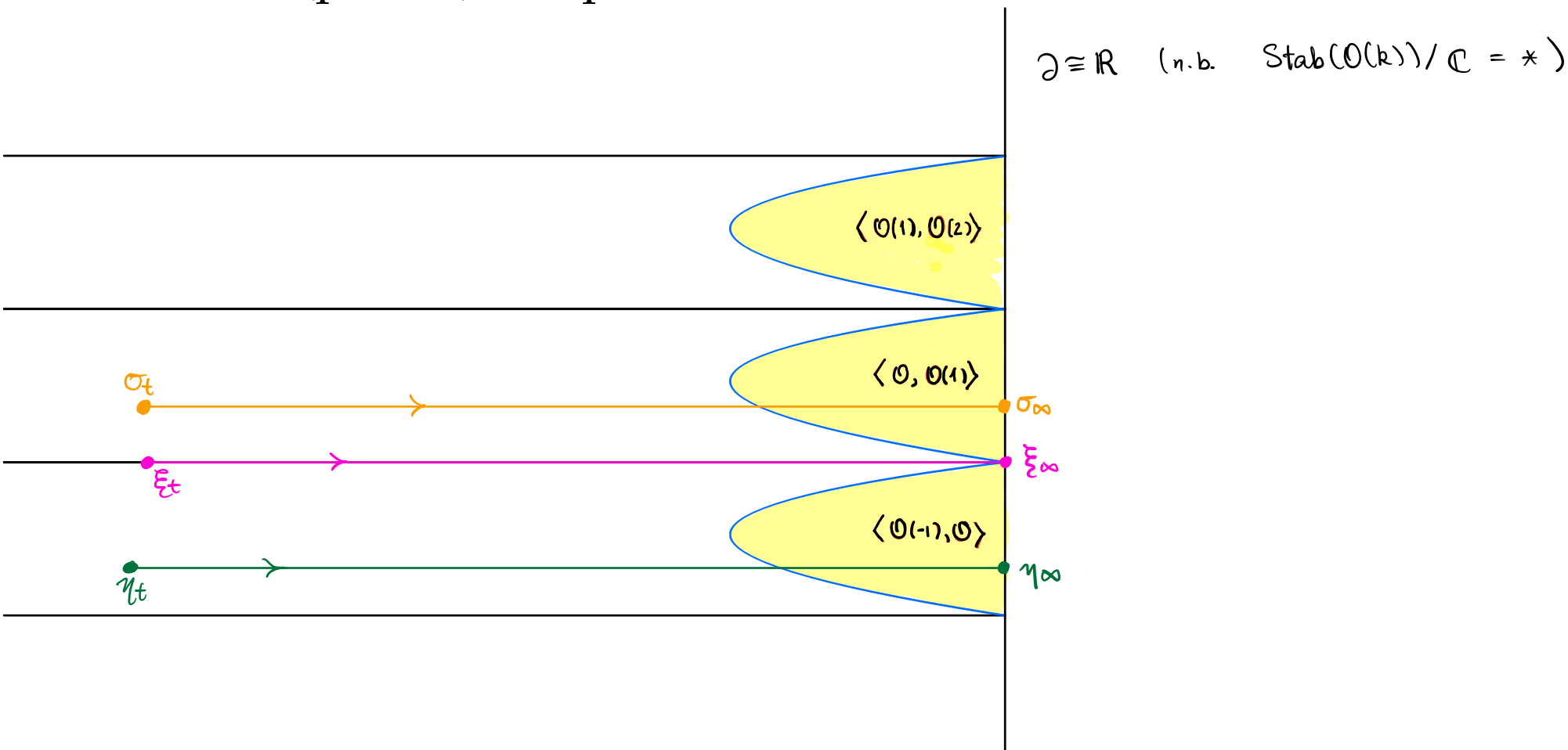
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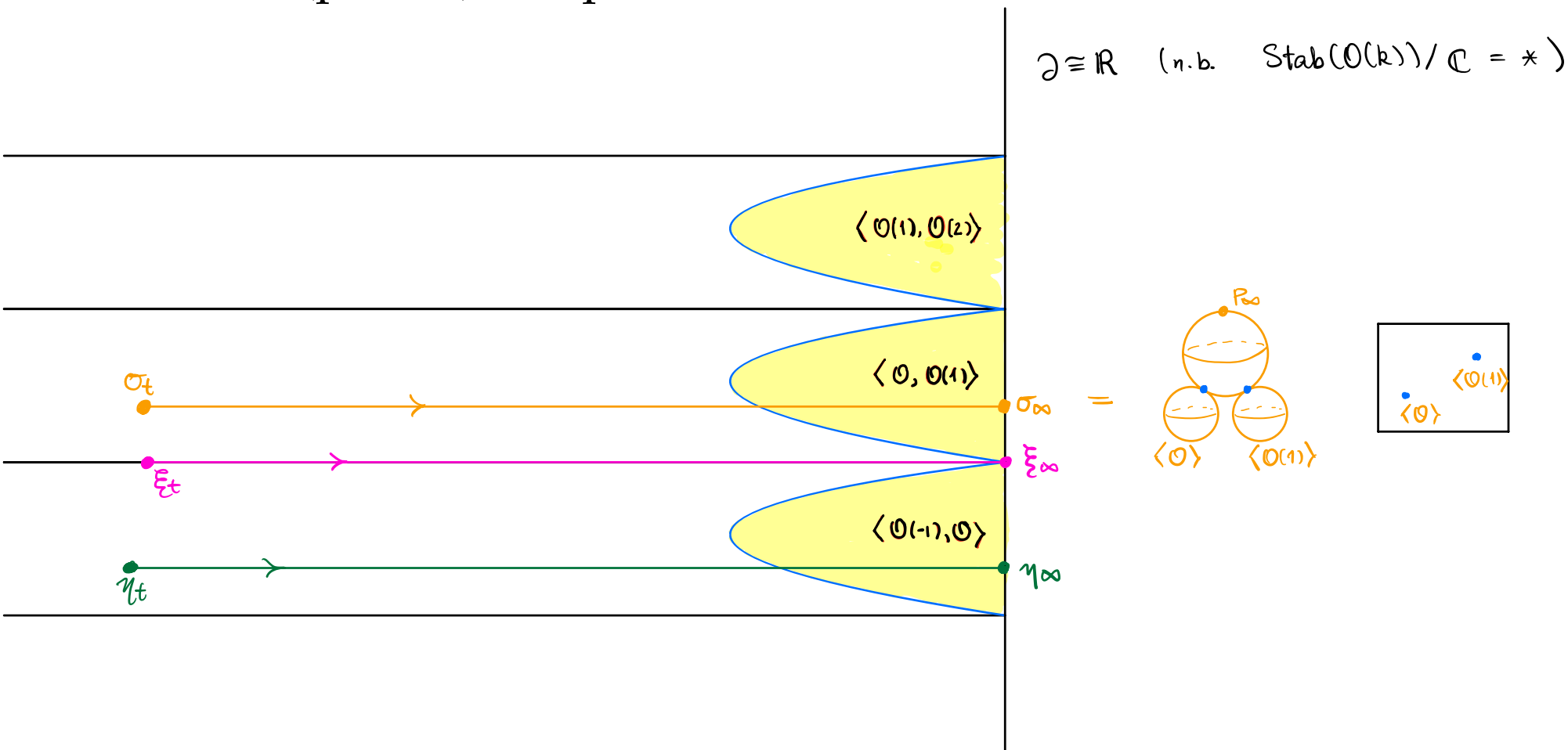
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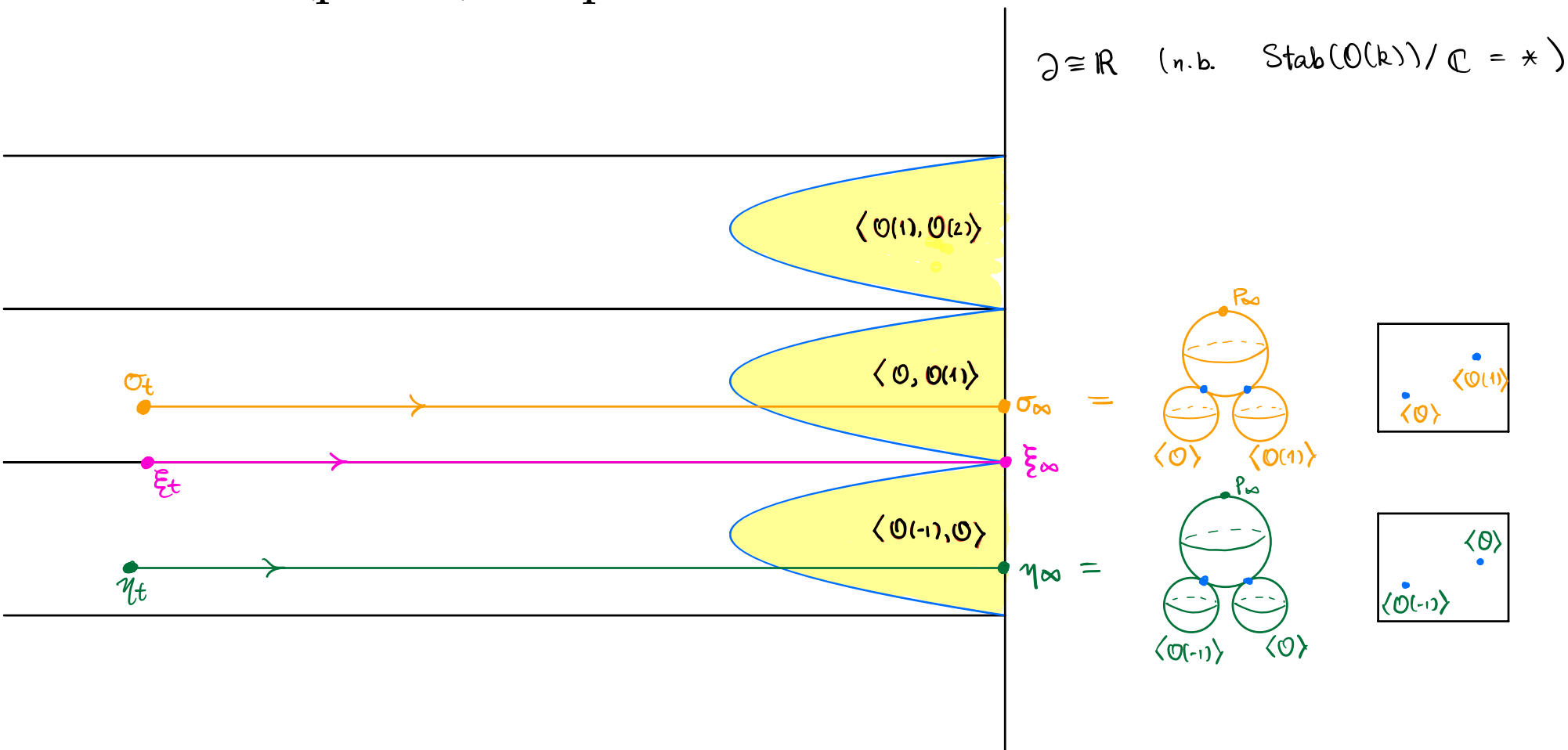
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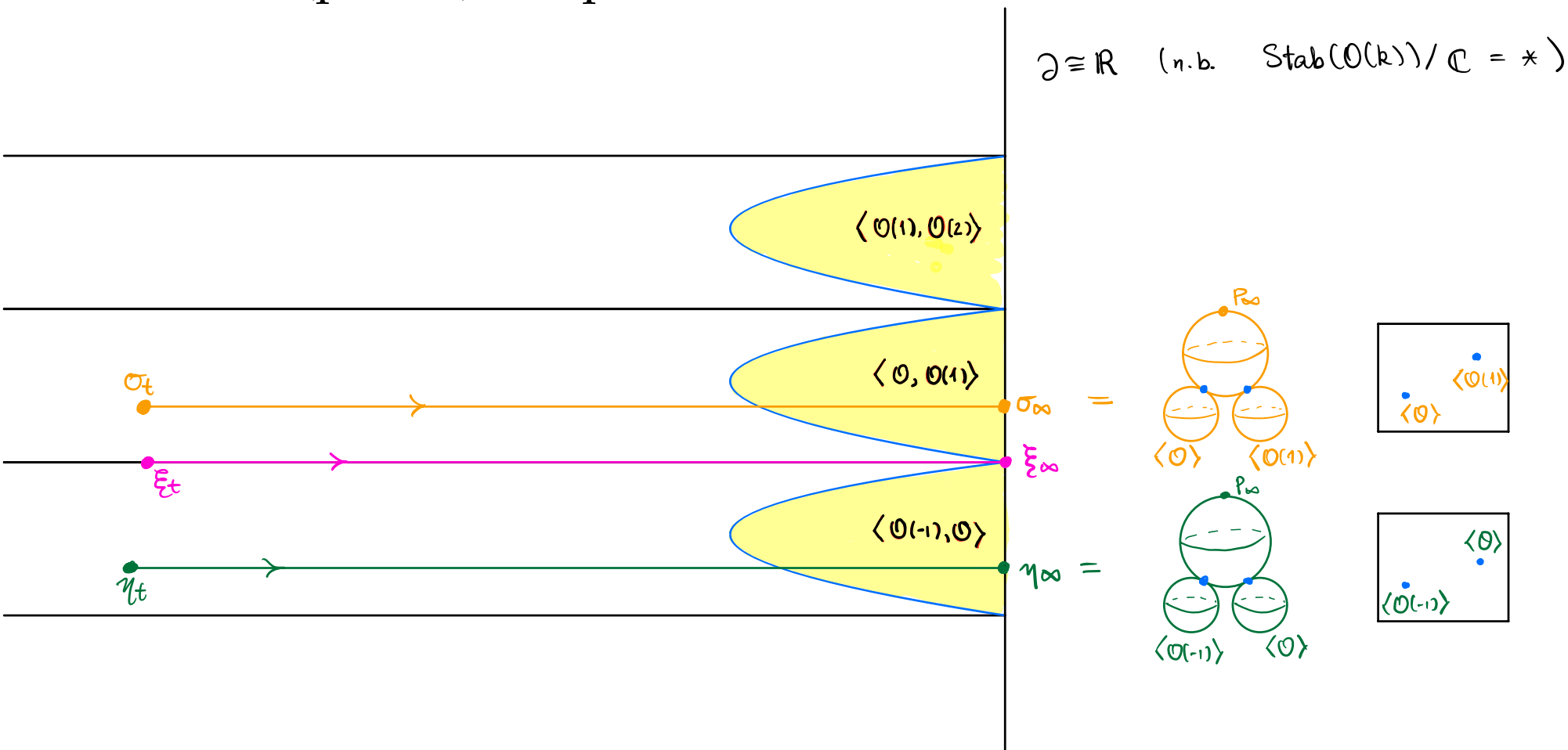
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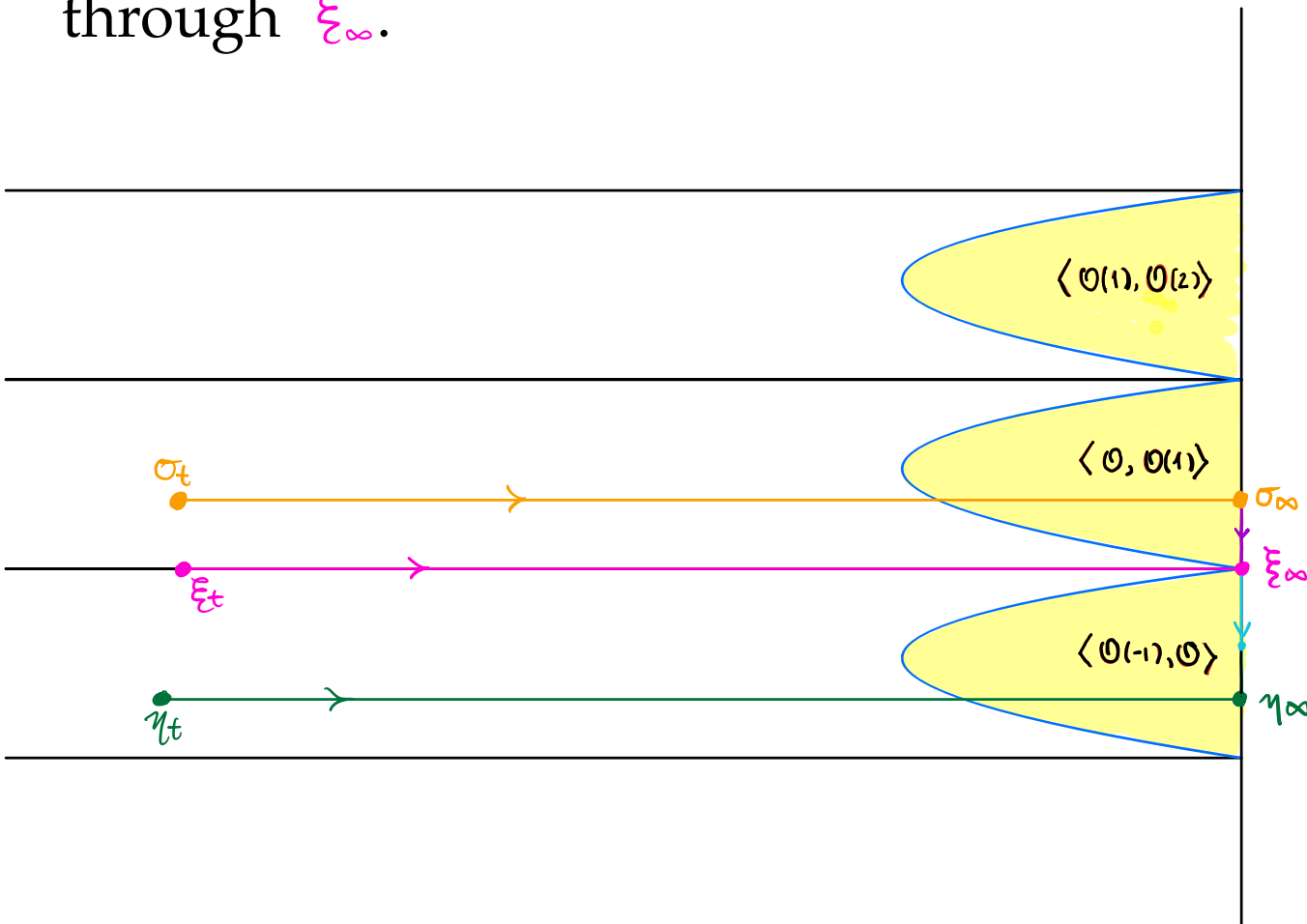
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Question: what is the limiting point of ξ_t as $t \rightarrow \infty$?

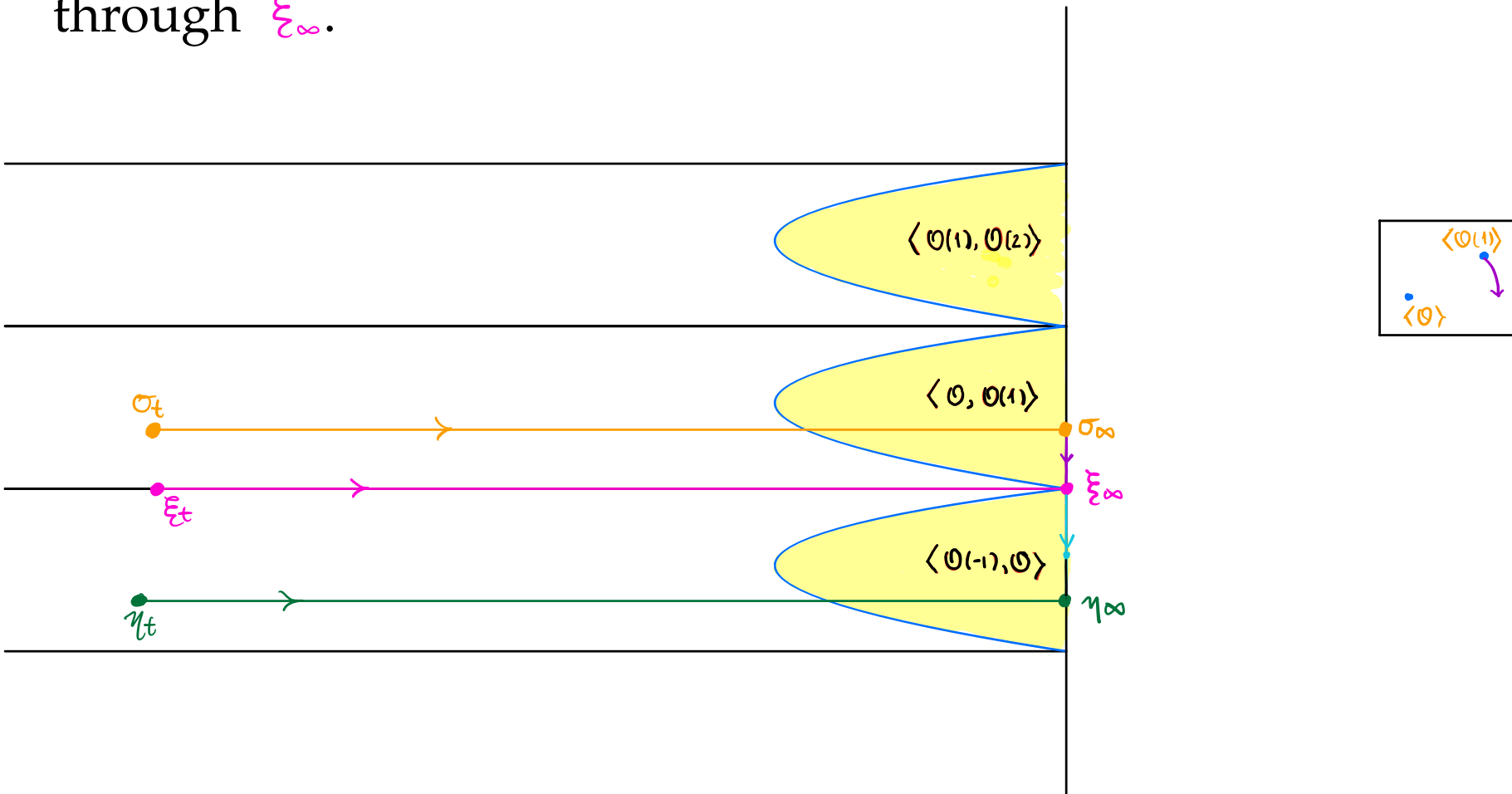
Path along the boundary

We consider a path along the boundary from σ_∞ to η_∞ , which passes through ξ_∞ .



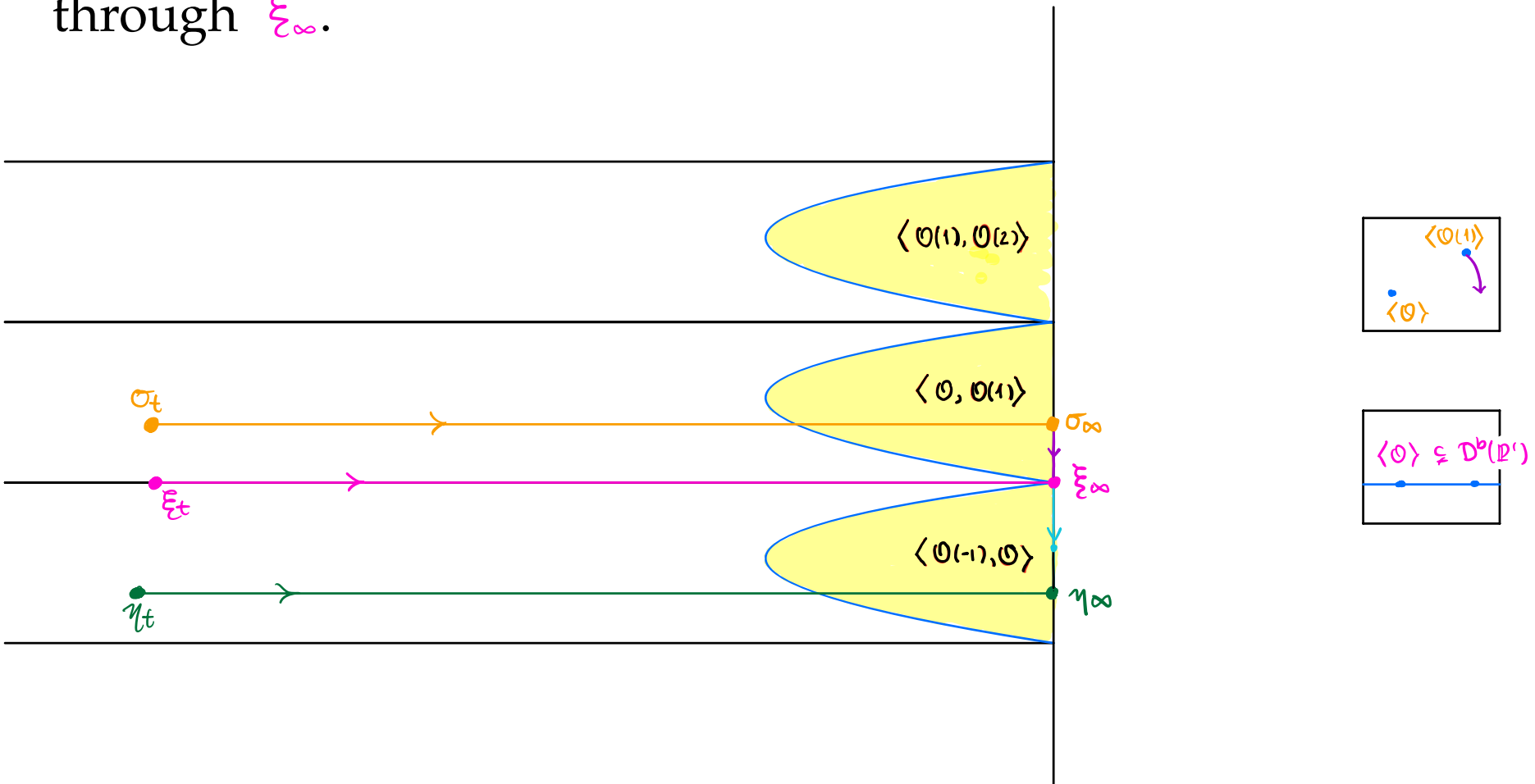
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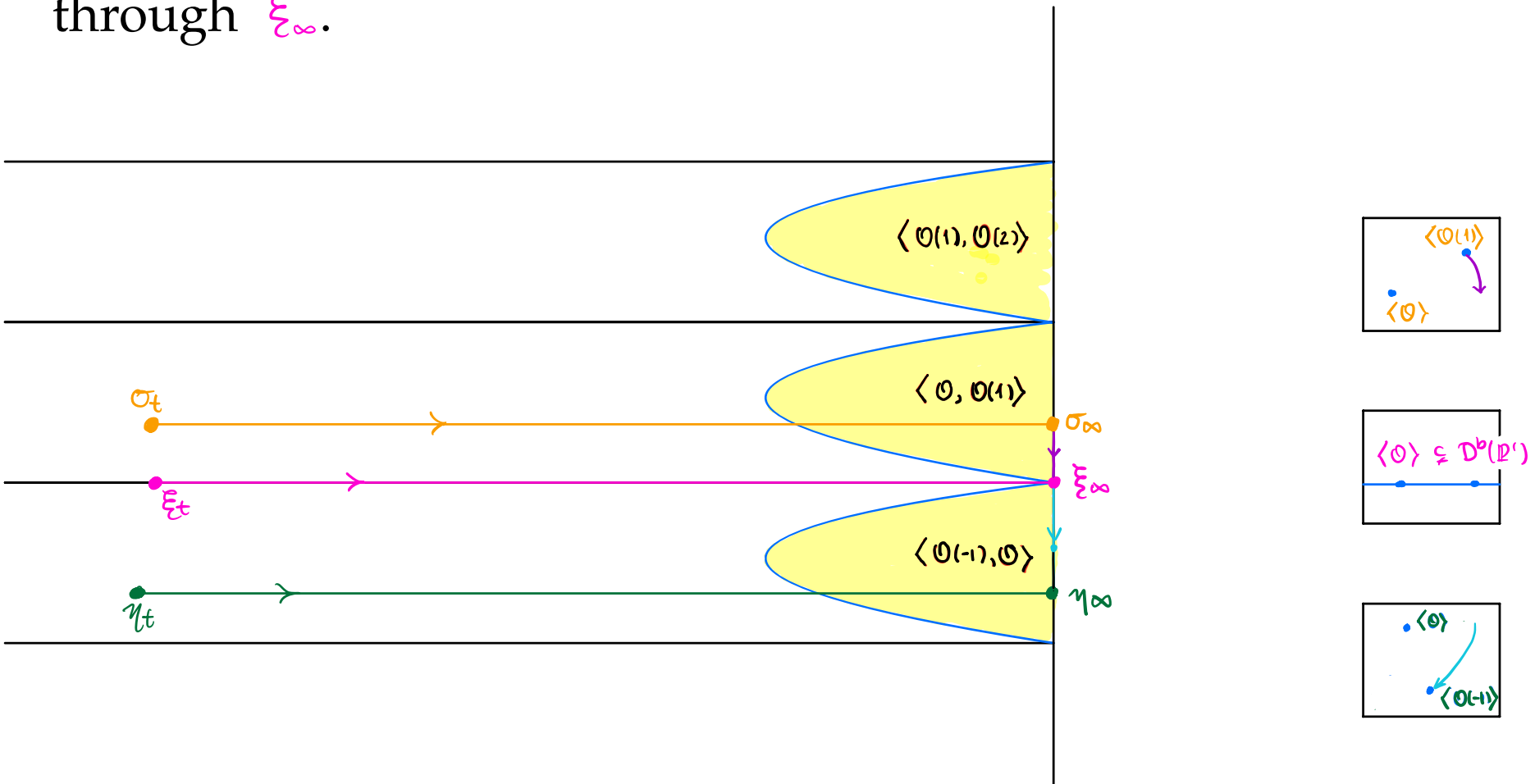
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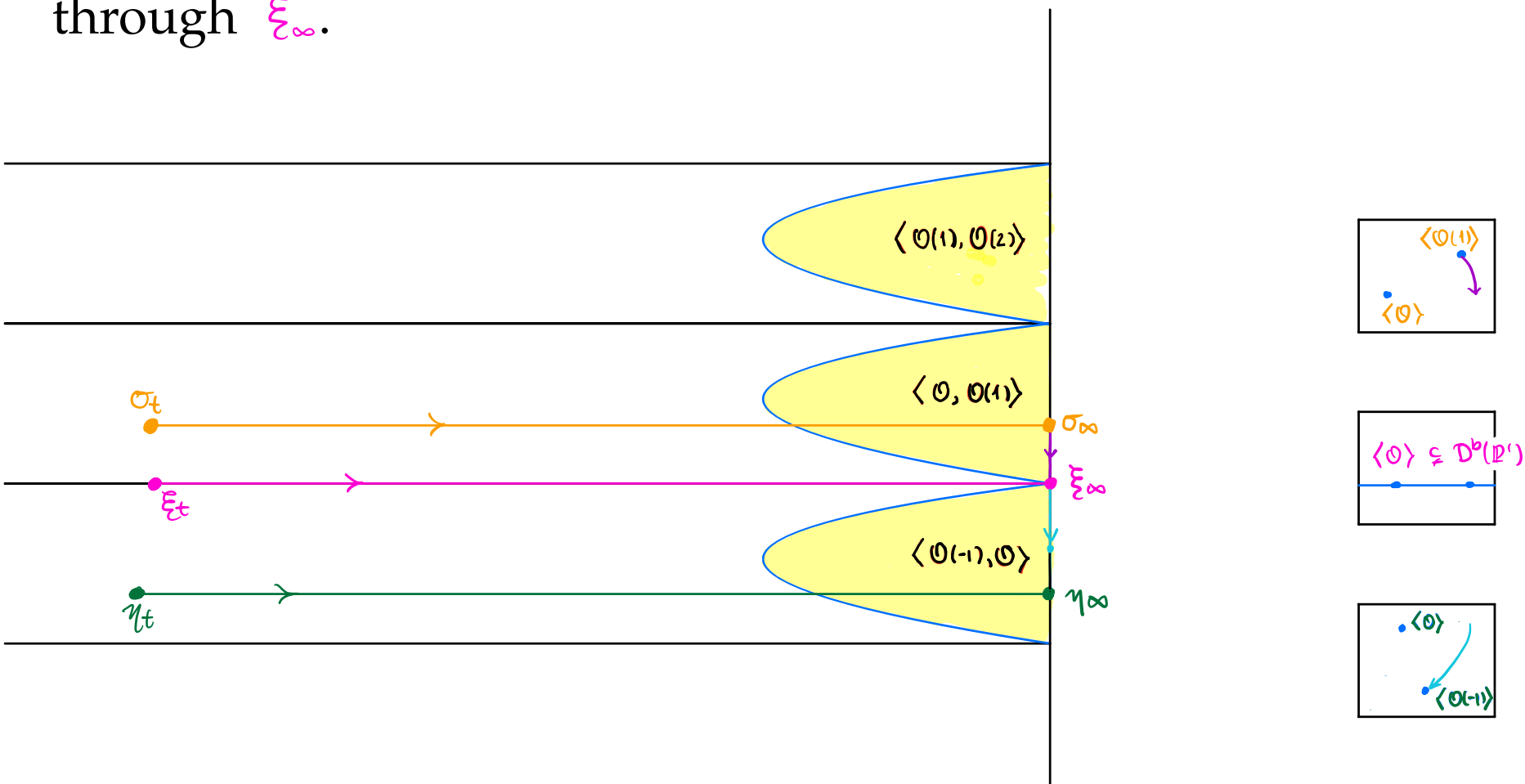
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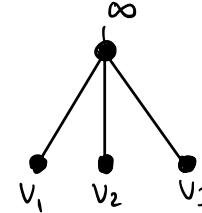


The boundary point $\lim_{t \rightarrow \infty} \xi_t$ is a *degenerate* semiorthogonal decomposition, i.e. an admissible *filtration* $0 \subsetneq \langle \mathcal{O} \rangle \subsetneq D_{\text{coh}}^b(\mathbf{P}^1)$.

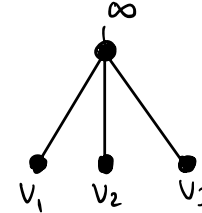
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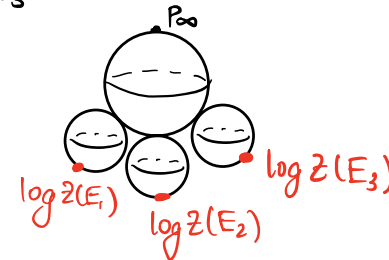


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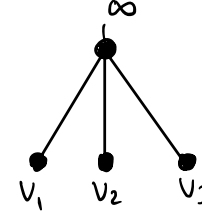
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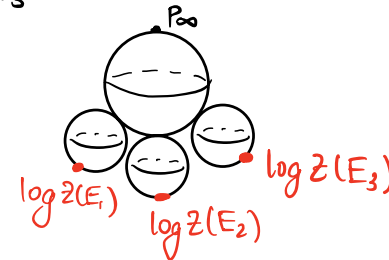
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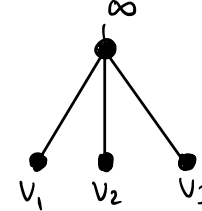
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where E_1, E_2, E_3 is obtained by mutation along b .

Proposition (Informal)

Connected components of strata in $\partial\mathcal{A}\text{Stab}$ correspond to equivalence classes of SODs up to mutation.

This gives us a revised:

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Given $\sigma_0, \tau_0 \in \text{Stab}(X)/\mathbf{C}$ and corresponding paths σ_t and τ_t , one hopes σ_t and τ_t converge to points in the same connected component of $\partial\mathcal{A}\text{Stab}(X)$, giving a canonical mutation class of SOD for $D_{\text{coh}}^b(X)$.

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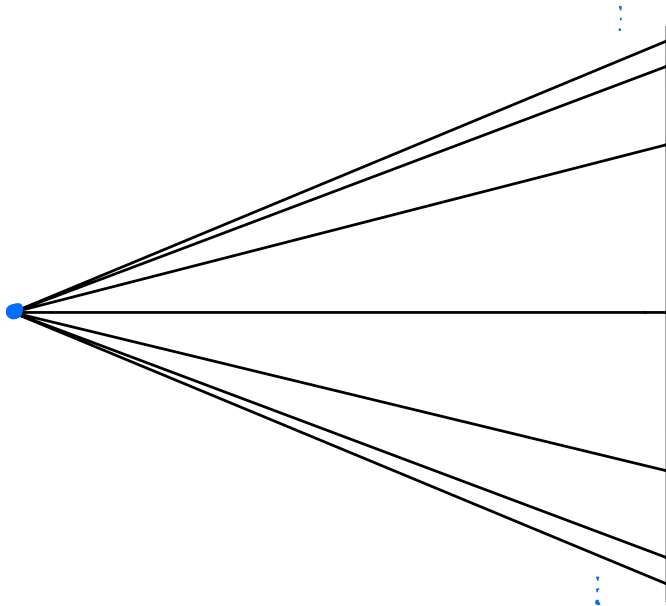
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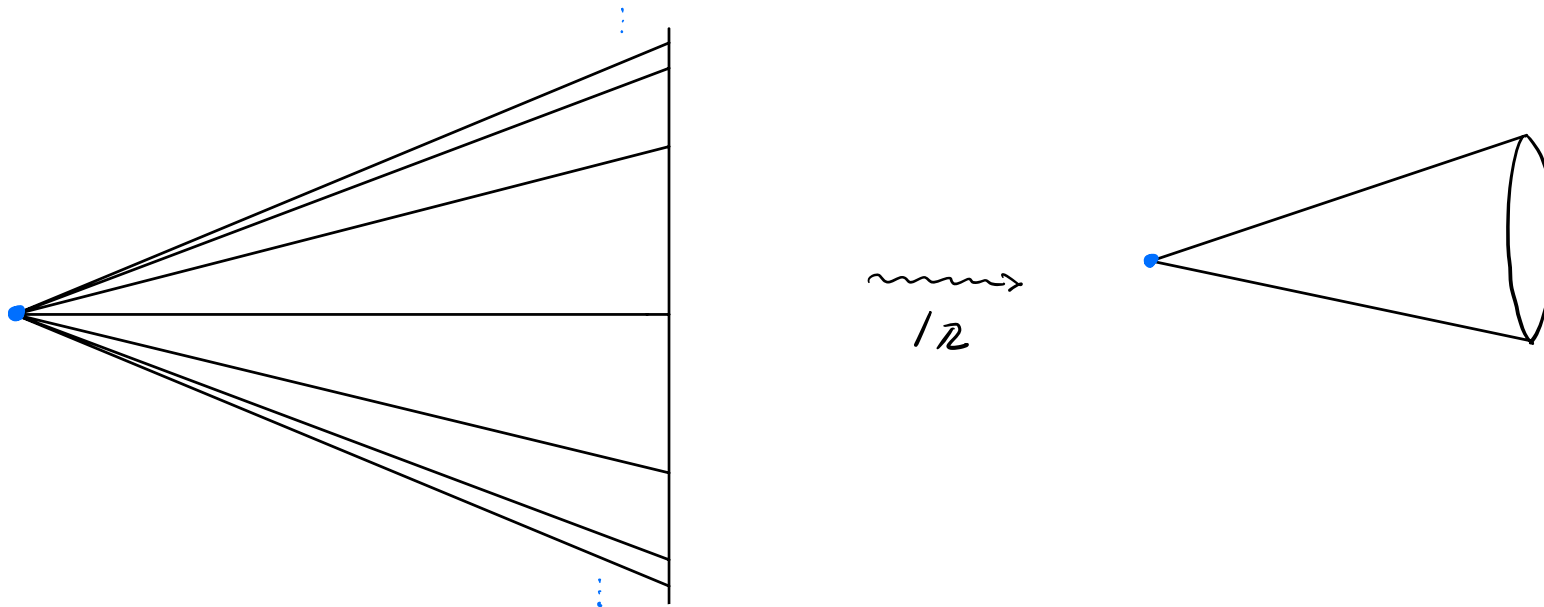


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